In …

```
(* Settings *)
SetOptions[
    Plot3D,
    Background -> White,
    BoxStyle -> Black,
    AxesStyle -> Black,
    LabelStyle -> Black,
    TicksStyle -> Black
];

SetOptions[
    ParametricPlot,
    Background -> White,
    AxesStyle -> Black,
    LabelStyle -> Black,
    TicksStyle -> Black
];
```

Practical-1

Solution of Cauchy problem for First-Order PDE

Problem-1: To obtain the solution of the linear equation

$$u_x - u_y = 1$$

with the Cauchy data $u(x, 0) = x^2$.

Solution:

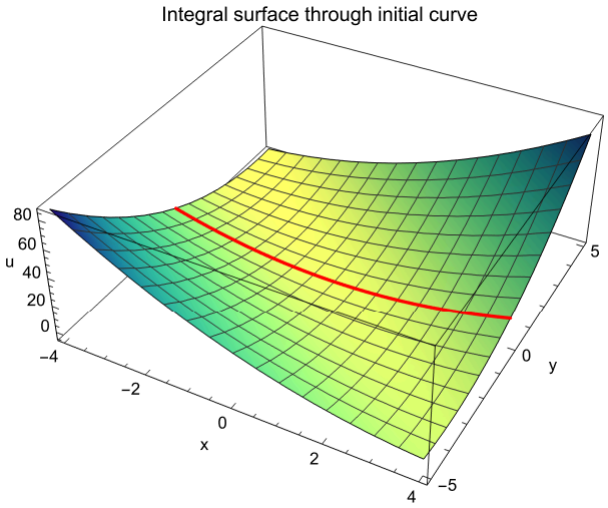
In …

```
Clear[x, y, u, t,
    pde, initialCondition,
    solution,
    surface, initialCurve
]

pde = D[u[x, y], x] - D[u[x, y], y] == 1

initialCondition = u[x, 0] == x^2;

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

surface = Plot3D[
    u[x, y] /. solution, {x, -4, 4}, {y, -5, 5},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Integral surface through initial curve",
    ColorFunction -> (ColorData["BlueGreenYellow"][1 - #3] &),
    ColorFunctionScaling -> True
];

initialCurve = ParametricPlot3D[
    {t, 0, t^2}, {t, -4, 4},
    PlotStyle -> {Thick, Red}
];

Show[surface, initialCurve]
```

Out…

$$-u^{(0,1)}[x, y] + u^{(1,0)}[x, y] == 1$$

$$\{\{u[x, y] \to x^2 - y + 2\,x\,y + y^2\}\}$$



Result: The solution of the Cauchy problem for the given PDE is obtained in the output and visualized as an integral surface.

Problem-2: To obtain the solution of the linear equation

$$u(x+y)\,u_x + u(x-y)\,u_y = x^2 + y^2$$

with the Cauchy data $u(x, 2x) = 0$.

Solution:

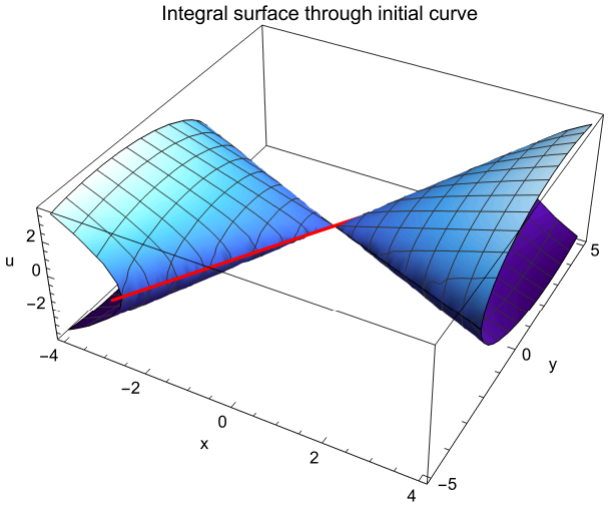
In …

```
Clear[x, y, u, t,
    pde, initialCondition,
    solution,
    surface, initialCurve
]

pde = u[x, y]*(x + y)*D[u[x, y], x] + u[x, y]*(x - y)*D[u[x, y], y] == x^2 + y^2

initialCondition = u[x, 2 x] == 0;

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

surface = Plot3D[
    u[x, y] /. solution, {x, -4, 4}, {y, -5, 5},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Integral surface through initial curve",
    ColorFunction -> (ColorData["DeepSeaColors"]),
    ColorFunctionScaling -> True
];

initialCurve = ParametricPlot3D[
    {t, 2*t, 0}, {t, -4, 4},
    PlotStyle -> {Thick, Red}
];

Show[surface, initialCurve]
```

Out…

$$(x-y)\,u[x,y]\,u^{(0,1)}[x,y]+(x+y)\,u[x,y]\,u^{(1,0)}[x,y]==x^2+y^2$$

$$\left\{\left\{u[x,y]\to-\sqrt{\frac{2}{7}}\sqrt{2x^2+3xy-2y^2}\right\},\left\{u[x,y]\to\sqrt{\frac{2}{7}}\sqrt{2x^2+3xy-2y^2}\right\},\left\{u[x,y]\to-\sqrt{\frac{2}{7}}\sqrt{2x^2+3xy-2y^2}\right\},\left\{u[x,y]\to\sqrt{\frac{2}{7}}\sqrt{2x^2+3xy-2y^2}\right\}\right\}$$



Result: The solution of the Cauchy problem for the given PDE is obtained in the output and visualized as an integral surface.

Problem-3: To obtain the solution of the linear equation

$$u_x + u_y = u^2$$

(a) with the Cauchy data $x + y = 0$, $u = 1$.
(b) with the Cauchy data $u(x, 0) = \tanh(x)$.

Solution:

(a)

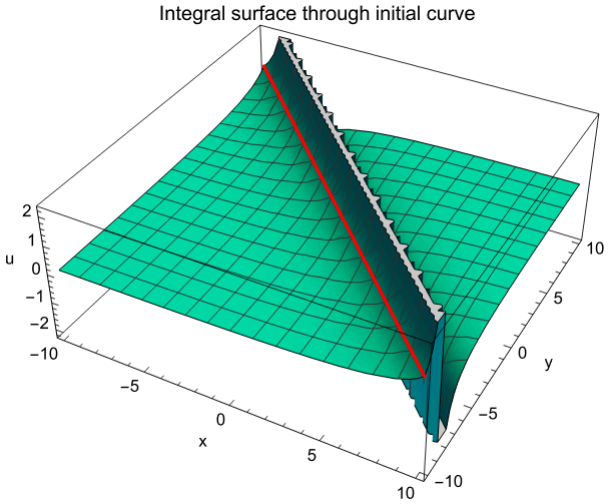
In …

```
Clear[x, y, u, t,
    pde, initialCondition,
    solution,
    surface, initialCurve
]

pde = D[u[x, y], x] + D[u[x, y], y] == (u[x, y])^2

(* x + y = 0, u = 1 => y = -x, u = 1 => u[x, -x] = 1 *)
initialCondition = u[x, -x] == 1;

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

surface = Plot3D[
    u[x, y] /. solution, {x, -10, 10}, {y, -10, 10},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Integral surface through initial curve",
    PlotStyle -> Cyan
];

initialCurve = ParametricPlot3D[
    {t, -t, 1}, {t, -10, 10},
    PlotStyle -> {Thick, Red}
];

Show[surface, initialCurve]
```

Out…

$$u^{(0,1)}[x, y] + u^{(1,0)}[x, y] == u[x, y]^2$$

$$\left\{\left\{u[x, y] \to -\frac{2}{-2 + x + y}\right\}\right\}$$



(b)

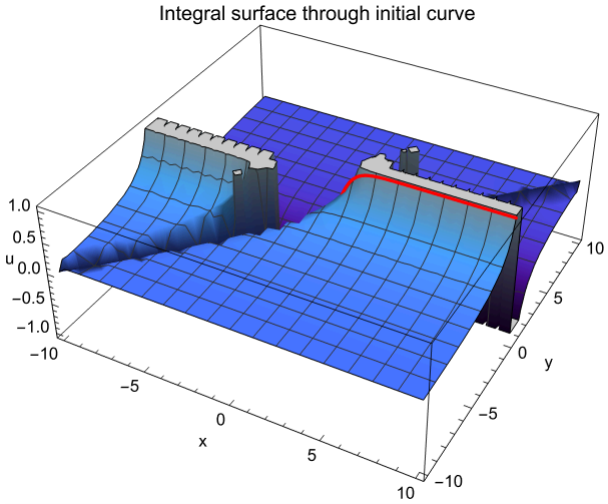
In …

```
Clear[x, y, u, t,
    pde, initialCondition,
    solution,
    surface, initialCurve
]

pde = D[u[x, y], x] + D[u[x, y], y] == (u[x, y])^2

initialCondition = u[x, 0] == Tanh[x];

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

surface = Plot3D[
    u[x, y] /. solution, {x, -10, 10}, {y, -10, 10},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Integral surface through initial curve",
    ColorFunction -> (ColorData["DeepSeaColors"]),
    ColorFunctionScaling -> True
];

initialCurve = ParametricPlot3D[
    {t, 0, Tanh[t]}, {t, -10, 10},
    PlotStyle -> {Thick, Red}
];

Show[surface, initialCurve]
```

Out…

```
u^(0,1)[x, y] + u^(1,0)[x, y] == u[x, y]^2
```

$$\left\{\left\{u[x, y] \to \frac{1}{-y + \text{Coth}[x - y]}\right\}\right\}$$



Result: The solution of the Cauchy problem for the given PDE is obtained in the output and visualized as an integral surface.

Practical-2

Plotting the Characteristics for the First-Order PDE

Problem-1: To find the characteristics of the equation

$$(u - y)\, u_x + y\, u_y = x + y$$

and plot them.

Solution:

The auxiliary equations of the given PDE is

$$\frac{dx}{u - y} = \frac{dy}{y} = \frac{du}{x + y} \tag{1}$$

We have,

$$\Rightarrow \frac{dx + du}{(u - y) + (x + y)} = \frac{dy}{y}$$

$$\Rightarrow \frac{d(x + u)}{x + u} = \frac{dy}{y}$$

$$\Rightarrow \int \frac{d(x + u)}{x + u} = \int \frac{dy}{y}$$

$$\Rightarrow \log(x + u) = \log(y) + \log(c_1)$$

$$\Rightarrow \frac{x + u}{y} = c_1$$

$$\therefore\ u = y\, c_1 - x \tag{2}$$

Here, c_1 is a constant.

Again, we have from Eq.(1),

$$\Rightarrow \frac{dx + dy}{(u - y) + y} = \frac{du}{x + y}$$

$$\Rightarrow \frac{d(x + y)}{u} = \frac{du}{x + y}$$

$$\Rightarrow (x + y)\, d(x + y) = u\, du$$

$$\Rightarrow \int (x + y)\, d(x + y) = \int u\, du$$

$$\Rightarrow \frac{(x + y)^2}{2} = \frac{u^2}{2} + \frac{c_2}{2}$$

$$\Rightarrow (x + y)^2 - u^2 = c_2$$

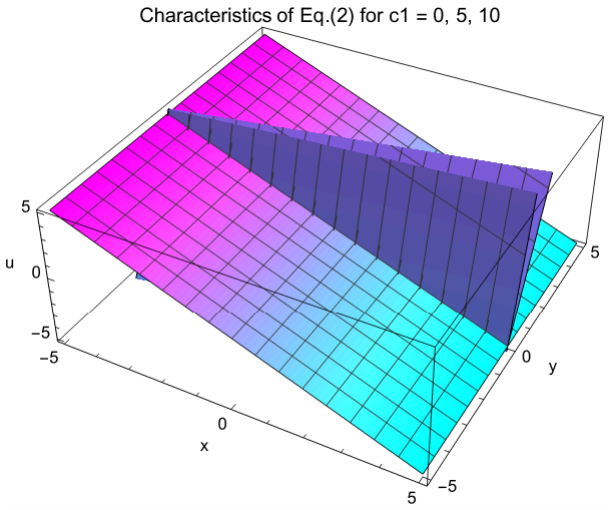
$$\therefore\ u = \sqrt{(x + y)^2 - c_2} \tag{3}$$

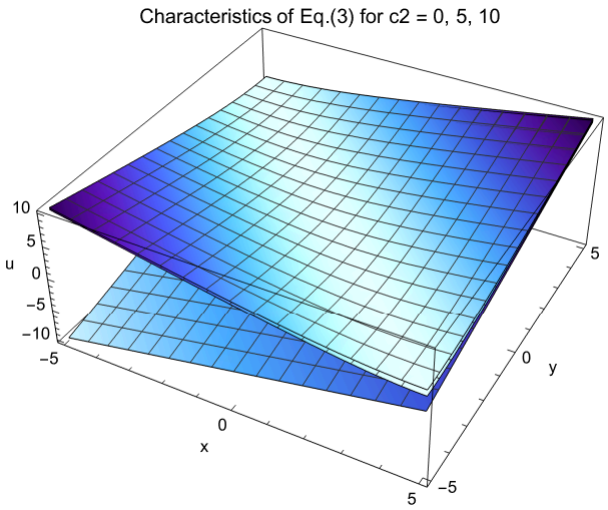
Here, c_2 is a constant.

In …

```
Clear[f0, f1, f2, h0, h1, h2]

(* for c1 = 0, 5, 10 *)
f0 = Plot3D[-x, {x, -5, 5}, {y, -5, 5},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Characteristics of Eq.(2) for c1 = 0, 5, 10",
    ColorFunction -> (Blend[{Cyan, Magenta}, #3] &),
    ColorFunctionScaling -> True
];
f1 = Plot3D[5*y - x, {x, -5, 5}, {y, -5, 5},
    ColorFunction -> (Blend[{Cyan, Magenta}, #3] &),
    ColorFunctionScaling -> True
];
f2 = Plot3D[10*y - x, {x, -5, 5}, {y, -5, 5},
    ColorFunction -> (Blend[{Cyan, Magenta}, #3] &),
    ColorFunctionScaling -> True
];

Show[f0, f1, f2]

(* for c2= 0, 5, 10 *)
h0 = Plot3D[x + y, {x, -5, 5}, {y, -5, 5},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Characteristics of Eq.(3) for c2 = 0, 5, 10",
    ColorFunction -> (ColorData["DeepSeaColors"][1 - #3] &),
    ColorFunctionScaling -> True
];
h1 = Plot3D[Sqrt[(x + y)^2 + 5], {x, -5, 5}, {y, -5, 5},
    ColorFunction -> (ColorData["DeepSeaColors"][1 - #3] &),
    ColorFunctionScaling -> True
];
h2 = Plot3D[Sqrt[(x + y)^2 + 10], {x, -5, 5}, {y, -5, 5},
    ColorFunction -> (ColorData["DeepSeaColors"][1 - #3] &),
    ColorFunctionScaling -> True
];

Show[h0, h1, h2]
```

Out…





Result: The characteristics of the given PDE are plotted in the output.

Practical-3

Plotting the Integral Surfaces of a given First-Order PDE with Initial Data

Problem-1: To find the solution of the equation

$$u_x + y\,u_y = 0$$

with the Cauchy data $u(0, y) = 4\,y$.
Plot the integral surface with in the range $(x, -4, 4)$ & $(y, -5, 5)$.

Solution:

In …

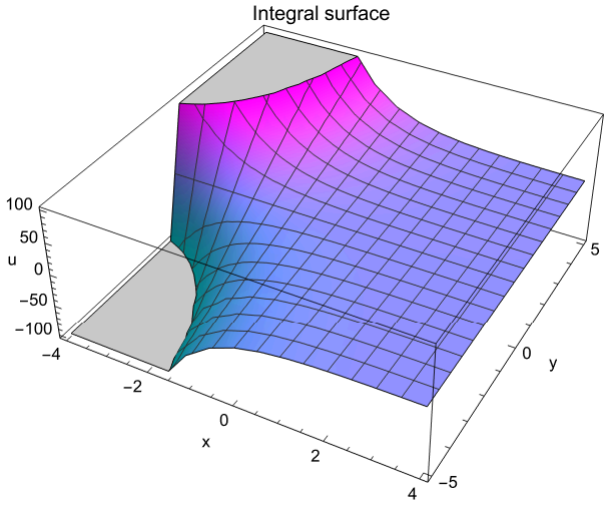
```
Clear[x, y, u, t,
    pde, initialCondition,
    solution,
    surface, initialCurve
]

pde = D[u[x, y], x] + y*D[u[x, y], y] == 0

initialCondition = u[0, y] == 4*y;

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

Plot3D[
    u[x, y] /. solution, {x, -4, 4}, {y, -5, 5},
```

```
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Integral surface",
    ColorFunction -> (Blend[{Cyan, Magenta}, #3] &),
    ColorFunctionScaling -> True
]
```

Out…

$y\,u^{(0,1)}[x, y] + u^{(1,0)}[x, y] == 0$

$\{\{u[x, y] \to 4\,e^{-x}\,y\}\}$



Result: The solution of the Cauchy problem for the given PDE is obtained in the output and visualized as an integral surface.

Problem-2: To obtain the solution of the equation

$$u_x + u\,u_y = 1$$

with the Cauchy data $u(0, y) = 4\,y$.
Plot the integral surface with in the range $(x, -4, 4)$ & $(y, -5, 5)$.

Solution:

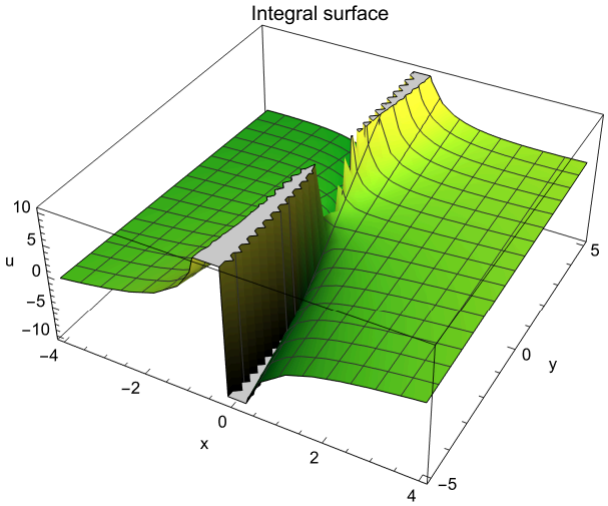
In …

```
Clear[x, y, u, t,
    pde, initialCondition,
    solution,
    surface, initialCurve
]

pde = D[u[x, y], x] + u[x, y]*D[u[x, y], y] == 1

initialCondition = u[0, y] == 4*y;

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

Plot3D[
    u[x, y] /. solution, {x, -4, 4}, {y, -5, 5},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Integral surface",
    ColorFunction -> "AvocadoColors",
    ColorFunctionScaling -> True
]
```

Out…

$$u[x, y]\, u^{(0,1)}[x, y] + u^{(1,0)}[x, y] == 1$$

$$\left\{\left\{u[x, y] \to \frac{x + 2x^2 + 4y}{1 + 4x}\right\}\right\}$$



Result: The solution of the Cauchy problem for the given PDE is obtained in the output and visualized as an integral surface.

Problem-3: To obtain the solution of the equation

$$u_x + u_y = 1$$

with the Cauchy data $u(x, 2x) = x^3$.
Plot the integral surface with in the range $(x, -4, 4)$ & $(y, -5, 5)$.

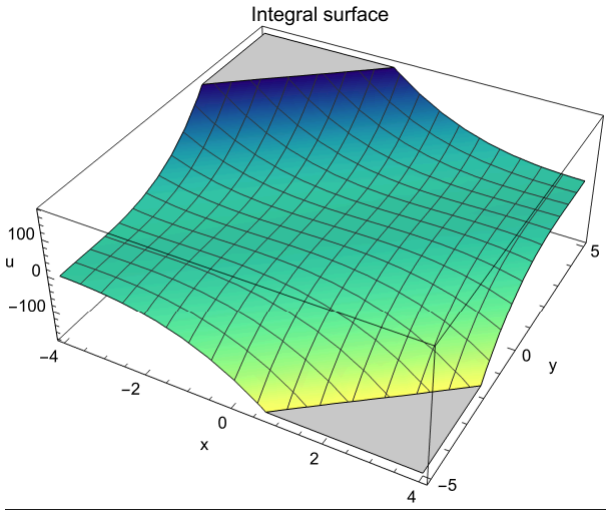
Solution:

In …

```
Clear[x, y, u, t,
    pde, initialCondition,
    solution,
    surface, initialCurve
]

pde = D[u[x, y], x] + D[u[x, y], y] == 1

initialCondition = u[x, 2 x] == x^3;

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

Plot3D[
    u[x, y] /. solution, {x, -4, 4}, {y, -5, 5},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Integral surface",
    ColorFunction -> (ColorData["BlueGreenYellow"][1 - #3] &),
    ColorFunctionScaling -> True
]
```

Out…

```
u^(0,1)[x, y] + u^(1,0)[x, y] == 1
{{u[x, y] → 2 x - x^3 - y + 3 x^2 y - 3 x y^2 + y^3}}
```



Result: The solution of the Cauchy problem for the given PDE is obtained in the output and visualized as an integral surface.